

Problem 1

Solution:

$$f(2) = 3$$

$$f(-2) = -1$$

$$-f(2) = -3$$

$$f\left(\frac{1}{2}\right) = \frac{3}{2}$$

$$\frac{1}{f(2)} = \frac{1}{3}$$

$$f(a+b) = a+b+1$$

$$f(a) + f(b) = a+b+2$$

$$f(a)f(b) = (a+1)(b+1)$$

□

Problem 2

Solution:

$$f(2) + g(2) = 2$$

$$f(2) - g(2) = 4$$

$$f(2)g(2) = -3$$

$$\frac{f(2)}{g(2)} = -3$$

$$f(g(2)) = 0$$

$$g(f(2)) = -2$$

$$f(a) + g(-a) = 2 + 2a$$

$$f(t)g(-t) = (1+t)^2$$

□

Problem 3

Solution:

$$\varphi(0) = 4$$

$$\varphi(1) = 2$$

$$\varphi(2) = 2$$

$$\varphi(3) = 2$$

$$\varphi(-1) = 6$$

$$\varphi(-2) = 8$$

$$\varphi(t+2) = \varphi(t) \implies |t-1| + |t+1| = |t-3| + |t-1| \implies |t+1| = |t-3|.$$

There are 4 total cases here

- $t+1 = t-3$. No solutions.
- $t+1 = -(t-3)$. Solving yields $t = 1$.
- $-(t+1) = t-3$. Equivalent to second case.
- $-(t+1) = -(t-3)$. Equivalent to first case.

Hence, $t = 1$.

□

Problem 4

Solution:

(a) $f(-x) = (-x)^2 = x^2 = f(x)$. True for all real x .

(b) $f(y) - f(x) = y^2 - x^2 = (y+x)(y-x)$. True for all real x, y .

(c) $f(x+h) - f(x) = (x+h)^2 - x^2 = x^2 + 2xh + h^2 - x^2 = 2xh + h^2$. True for all real x, h .

(d) $f(2y) = (2y)^2 = 4y^2 = 4(y^2) = 4f(y)$. True for all real y .

(e) $f(t^2) = (t^2)^2 = (f(t))^2$. True for all real t .

(f) $\sqrt{f(a)} = \sqrt{a^2} = |a|$ since square root is (by convention) the positive root. True for all real a .

□

Problem 5

Solution:

(a) $g(-x) = \sqrt{4 - (-x)^2} = \sqrt{4 - x^2} = g(x)$. True for all real $|x| \leq 2$.

(b) $g(2y) = \sqrt{4 - (2y)^2} = \sqrt{4 - 4y^2} = \sqrt{4}\sqrt{1 - y^2} = 2\sqrt{1 - y^2}$. True for all real $|y| \leq 1$.

(c)

$$\begin{aligned} g\left(\frac{1}{t}\right) &= \sqrt{4 - \left(\frac{1}{t}\right)^2} \\ &= \sqrt{4 - \frac{1}{t^2}} \\ &= \sqrt{\frac{4t^2 - 1}{t^2}} \\ &= \frac{\sqrt{4t^2 - 1}}{\sqrt{t^2}} \\ &= \frac{\sqrt{4t^2 - 1}}{|t|} \end{aligned}$$

Now, $t \neq 0$, but we also need

$$4t^2 - 1 \geq 0 \implies t^2 - \frac{1}{4} = \left(t + \frac{1}{2}\right)\left(t - \frac{1}{2}\right) \geq 0$$

Hence, both operands must be ≥ 0 or ≤ 0 . There are 4 sub-cases.

- $t \leq -\frac{1}{2}$. Possible since $t + \frac{1}{2} \leq 0$ and $t - \frac{1}{2} \leq 0$.
- $t \geq \frac{1}{2}$. Possible since $t + \frac{1}{2} \geq 0$ and $t - \frac{1}{2} \geq 0$.
- $-\frac{1}{2} < t < 0$. Not possible since $0 < t + \frac{1}{2} < \frac{1}{2}$ but $-1 < t - \frac{1}{2} < -\frac{1}{2} < 0$
- $0 < t < \frac{1}{2}$. Not possible since $0 < \frac{1}{2} < t + \frac{1}{2} < 1$ but $-1 < t - \frac{1}{2} < 0$

Thus, the domain is $t \leq -\frac{1}{2}$ or $t \geq \frac{1}{2}$.

(d)

$$\begin{aligned} g(a - 2) &= \sqrt{4 - (a - 2)^2} \\ &= \sqrt{4 - (a^2 - 4a + 4)} \\ &= \sqrt{4a - a^2} \end{aligned}$$

We need $4a - a^2 \geq 0 \implies a(4 - a) \geq 0$. There are two cases. If $a(4 - a) = 0$, then $a = 0$ or $a = 4$. If $a(4 - a) > 0$ then either both a and $4 - a$ must be positive or both must be negative. There are three sub-cases here.

- $a > 4$. Not possible: $a > 0$ but $4 - a < 0$
- $0 < a < 4$. Possible: $a > 0$ and $4 - a > 0$
- $a < 0$. Not possible: $4 - a > 0$ but $a < 0$

True for $0 \leq a \leq 4$

(e)

$$\begin{aligned}
 g\left(\frac{s}{2}\right) &= \sqrt{4 - \left(\frac{s}{2}\right)^2} \\
 &= \sqrt{4 - \frac{s^2}{4}} \\
 &= \sqrt{\frac{16 - s^2}{4}} \\
 &= \frac{\sqrt{16 - s^2}}{\sqrt{4}} \\
 &= \frac{1}{2}\sqrt{16 - s^2}
 \end{aligned}$$

True for all real $|s| \leq 4$.

(f)

$$\begin{aligned}
 \frac{1}{2 + g(x)} &= \frac{1}{2 + \sqrt{4 - x^2}} \\
 &= \left(\frac{1}{2 + \sqrt{4 - x^2}}\right) \left(\frac{2 - \sqrt{4 - x^2}}{2 - \sqrt{4 - x^2}}\right) \\
 &= \frac{2 - \sqrt{4 - x^2}}{4 - (\sqrt{4 - x^2})^2} \\
 &= \frac{2 - \sqrt{4 - x^2}}{4 - (4 - x^2)} \\
 &= \frac{2 - \sqrt{4 - x^2}}{x^2} \\
 &= \frac{2 - g(x)}{x^2}
 \end{aligned}$$

True for all real $-2 \leq x < 0 \cup 0 < x \leq 2$.

□

Problem 6

Solution:

All graphs are omitted. Use <https://www.desmos.com/> instead.

(b) g is only defined on $0 \leq x \leq 1$

- $g(x) = 1$ if $0 \leq x \leq \frac{1}{2}$

- $g(x) = 2$ if $\frac{1}{2} < x \leq 1$

(c) h is only defined on $2 \leq x \leq 4$

- $h(x) = 1$ if $2 \leq x \leq 3$
- $h(x) = 2$ if $3 < x \leq 4$

(d) k is not defined anywhere. $k(x) = g(x) + h(x)$ but the domains of g and h don't intersect. □

Problem 7

Solution:

All graphs are omitted. Use <https://www.desmos.com/> instead.

The graphs of g, f intersect at the points (x, y) where $y = g(x) = f(x)$. Hence,

$$\begin{aligned} g(x) - f(x) &= 0 \\ &= x - x^3 \\ &= x(1 - x^2) \\ &= x(1 + x)(1 - x) \end{aligned}$$

Thus, $x = 0, -1, 1$ are possible x and the intersection points are $(0, 0), (-1, -1), (1, 1)$. □

Problem 8

Solution:

All graphs are omitted. Use <https://www.desmos.com/> instead.

The graphs of g, f intersect at the points (x, y) where $y = g(x) = f(x)$. Hence,

$$\begin{aligned} g(x) - f(x) &= 0 \\ &= (x^2 - 2) - (2x^2 + 4x + 1) \\ &= x^2 - 2 - 2x^2 - 4x - 1 \\ &= -x^2 - 4x - 3 \\ &= -(x^2 + 4x + 3) \\ &= -(x + 3)(x + 1) \end{aligned}$$

Thus, $x = -3, -1$ are possible x and the intersection points are $(-3, 7), (-1, -1)$. □

Problem 9

Solution:

(a) Observe that

$$f(0) = 0 = \sum_{k=0}^n c_k(0)^k = c_0(0)^0 + c_1(0)^1 + \cdots + c_n(0)^n = c_0 + 0 + \cdots + 0 = c_0$$

Therefore,

$$f(x) = c_0x^0 + \sum_{k=1}^n c_kx^k = 0x^0 + \sum_{k=1}^n c_kx^k = \sum_{k=1}^n c_kx^k$$

Now, let g be an $n - 1$ -degree polynomial

$$g(x) = \sum_{k=0}^{n-1} d_kx^k$$

where $d_k = c_{k+1}$. Then, observe that

$$\begin{aligned} x(g(x)) &= x \left(\sum_{k=0}^{n-1} c_{k+1}x^k \right) \\ &= \sum_{k=0}^{n-1} c_{k+1}x^{k+1} \\ &= \sum_{k=1}^n c_kx^k \\ &= f(x) \end{aligned}$$

(b) Let a be an arbitrary fixed real. We will illustrate, by induction on n , that given an arbitrary n -degree polynomial $f_n(x) = \sum_{k=1}^n c_kx^k$, there always exists a n -degree polynomial $p_n(x)$ such that $p_n(x) = f_n(x + a)$.

For the base case, observe $f_0(x) = \sum_{k=0}^0 c_0x^k = c_0x^0 = c_0$. Then, $p_0(x) = f_0(x + a) = c_0$ is a zero-degree polynomial. For the inductive hypothesis, assume that, for every degree $0 \leq j \leq n - 1$ polynomial f there exists a degree- j polynomial $p_j(x)$ such that $p_j(x) = f_j(x + a)$.

Now, for the inductive step, we must show that $p_n(x) = f_n(x + a)$ is a degree- n polynomial.

Observe that

$$\begin{aligned}
 p_n(x) &= f_n(x+a) \\
 &= \sum_{k=0}^n c_k(x+a)^k \\
 &= \left(\sum_{k=0}^{n-1} c_k(x+a)^k \right) + c_n(x+a)^n \\
 &= f_{n-1}(x+a) + c_n(x+a)^n \\
 &= p_{n-1}(x) + c_n(x+a)^n && \text{(by inductive hypothesis)} \\
 &= \left(\sum_{k=0}^{n-1} d_k x^k \right) + c_n(x+a)^n && \text{(for some real } d_k) \\
 &= \left(\sum_{k=0}^{n-1} d_k x^k \right) + c_n \left(\sum_{k=0}^n \binom{n}{k} x^k a^{n-k} \right) && \text{(by binomial theorem)} \\
 &= \left(\sum_{k=0}^{n-1} d_k x^k \right) + c_n \left(\sum_{k=0}^{n-1} \binom{n}{k} x^k a^{n-k} \right) + c_n \binom{n}{n} x^n a^{n-n} \\
 &= \left(\sum_{k=0}^{n-1} x^k \left[d_k + c_n \binom{n}{k} a^{n-k} \right] \right) + c_n \binom{n}{n} x^n \\
 &= \left(\sum_{k=0}^{n-1} e_k x^k \right) + c_n x^n && (e_k = d_k + c_n \binom{n}{k} a^{n-k}) \\
 &= \sum_{k=0}^n e_k x^k && (e_n = c_n)
 \end{aligned}$$

Hence, we've shown that $p_n(x)$ can be written in the standard form of a degree- n polynomial.

(c) From part (b), we know that there exists a degree n polynomial $p(x)$ such that $p(x) = f(x+a)$. Now, observe that $p(0) = f(0+a) = f(a) = 0$. Since we also have $n \geq 1$, by part (a), there exists a degree $(n-1)$ polynomial $g(x)$ such that $p(x) = xg(x)$.

Observe that $p(x-a) = f((x-a)+a) = f(x)$ and $p(x-a) = (x-a)g(x-a)$. That is, $f(x) = (x-a)g(x-a)$.

Let $h(x) = g(x-a)$. By part (b) we know that h is also a polynomial of degree $n-1$ because $h(x+a) = g((x+a)-a) = g(x)$. Thus, we have shown that $f(x) = (x-a)h(x)$ for a suitable h .

(d) We use induction on the degree of the polynomial n . For the base case ($n=0$), observe that $f(x) = \sum_{k=0}^0 c_k x^k = c_0 x^0 = c_0$. Hence, if there is a real a such that $f(a) = 0$, we must have $c_0 = 0$. In other words, $f(x) = 0$ for all real x .

For the inductive hypothesis, assume that given a polynomial $f(x)$ of arbitrary fixed degree $m \geq 0$, if $f(x)$ has $m + 1$ distinct zeroes $a_1, a_2, \dots, a_m, a_{m+1}$, then $f(x) = 0$ for all real x .

For the inductive step, we must show that the same holds for an $f(x)$ of degree $m + 1$ with $m + 2$ distinct zeroes $a_1, \dots, a_{m+1}, a_{m+2}$. Since $f(a_{m+2}) = 0$ and $m \geq 0 \implies m + 1 \geq 1$, by part (c), there exists a degree $m = (m + 1) - 1$ polynomial $h(x)$ such that

$$f(x) = (x - a_{m+2})h(x)$$

Now, since $a_1, a_2, \dots, a_{m+1}$ are all zeroes of $f(x)$, $f(a_j) = (a_j - a_{m+2})h(a_j) = 0$ must be true for all $1 \leq j \leq m + 1$. And since all the zeroes are distinct, we know that $a_j - a_{m+2} \neq 0$ for all j . Therefore, we must have $h(a_j) = 0$ for all j , i.e. h has at least $m + 1$ zeroes $a_1, a_2, \dots, a_{m+1}$.

But by the inductive hypothesis, this means $h(x) = 0$ for all real x . Hence, we also have

$$f(x) = (x - a_{m+2})(0) = 0$$

(e) Let

$$\begin{aligned} h(x) &= g(x) - f(x) \\ &= \sum_{k=0}^m b_k x^k - \sum_{k=0}^n c_k x^k \\ &= \sum_{k=0}^m b_k x^k - \sum_{k=0}^n c_k x^k \\ &= \left[\sum_{k=0}^n b_k x^k + \sum_{k=n+1}^m b_k x^k \right] - \sum_{k=0}^n c_k x^k \\ &= \sum_{k=0}^n (b_k - c_k) x^k + \sum_{k=n+1}^m b_k x^k \\ &= \sum_{k=0}^m d_k x^k \end{aligned}$$

where $d_k = b_k - c_k$ for $0 \leq k \leq n$ and $d_k = b_k$ for $(n + 1) \leq k \leq m$. Hence $h(x)$ is a degree m polynomial. Now, since there are $m + 1$ distinct reals a_j such that $g(a_j) = f(a_j)$, observe that $h(a_j) = g(a_j) - f(a_j) = 0$. That is, $h(x)$ has $m + 1$ distinct real zeroes. But by part (d), this means that $h(x) = 0$ for all real x .

So $d_k = 0$ for all $0 \leq k \leq m$, which implies that $b_k - c_k = 0 \implies b_k = c_k$ for all $0 \leq k \leq n$.

Further, $b_k = 0$ for all $(n + 1) \leq k \leq m$ so $g(x) = \sum_{k=0}^m b_k x^k = \sum_{k=0}^n b_k x^k$. And since g is of degree m , $b_m \neq 0$. Hence, m cannot be greater than n (i.e. $m \leq n$) because otherwise, we must have $b_m = 0$. This, combined with the fact that $m \geq n$ gives us $m = n$.

Finally, $h(x) = 0 \implies g(x) = f(x)$ for all real x .

□

Problem 10

Solution:

Let p be a polynomial of degree at most 2, i.e. $p(x) = c_0 + c_1x + c_2x^2$ (c_k may be zero).

(a)

- $p(0) = c_0$ so $p(0) = 1 \implies c_0 = 1$
- $p(1) = c_0 + c_1 + c_2$ so $p(1) = 1 \implies 1 + (c_1 + c_2) = 1 \implies c_1 + c_2 = 0$
- $p(2) = c_0 + 2c_1 + 4c_2$ so

$$p(2) = 1 \implies 1 + (2c_1 + 4c_2) = 1 \implies 2(c_1 + c_2) + 2c_2 = 0 \implies c_2 = 0$$

But this in turn implies that $c_1 = 0$. So the only possible p is $p(x) = 1$.

(b)

- $p(0) = c_0$ so $p(0) = 1 \implies c_0 = 1$
- $p(1) = c_0 + c_1 + c_2$ so $p(1) = 1 \implies 1 + (c_1 + c_2) = 1 \implies c_1 + c_2 = 0$
- $p(2) = c_0 + 2c_1 + 4c_2$ so

$$p(2) = 2 \implies 1 + (2c_1 + 4c_2) = 2 \implies 2(c_1 + c_2) + 2c_2 = 1 \implies c_2 = \frac{1}{2}$$

But this in turn implies that $c_1 = -\frac{1}{2}$. So the only possible p is $p(x) = 1 - \frac{1}{2}x + \frac{1}{2}x^2$

(c)

- $p(0) = c_0$ so $p(0) = 1 \implies c_0 = 1$
- $p(1) = c_0 + c_1 + c_2$ so $p(1) = 1 \implies 1 + (c_1 + c_2) = 1 \implies c_1 + c_2 = 0$

Hence, any $p(x) = 1 + c_1x + c_2x^2$ where $c_1 + c_2 = 0$ satisfies the conditions.

(d)

- $p(0) = c_0$
- $p(1) = c_0 + c_1 + c_2$

so $p(0) = p(1) \implies c_1 + c_2 = 0$.

Hence, any $p(x) = c_0 + c_1x + c_2x^2$ where $c_1 + c_2 = 0$ works, i.e. c_0 can be any real.

□

Problem 11

Solution:

Let p be a polynomial of degree at most 2, i.e. $p(x) = c_0 + c_1x + c_2x^2$ (c_k may be zero).

(a)

$$\begin{aligned} p(x) = p(1-x) &\implies c_0 + c_1x + c_2x^2 = c_0 + c_1(1-x) + c_2(1-x)^2 \\ &\implies c_0 + c_1x + c_2x^2 = c_0 + c_1 - c_1x + c_2(1-2x+x^2) \\ &\implies c_0 + c_1x + c_2x^2 = c_0 + c_1 - c_1x + c_2 - 2c_2x + c_2x^2 \\ &\implies c_0 + c_1x + c_2x^2 = (c_0 + c_1 + c_2) - (c_1 + 2c_2)x + c_2x^2 \end{aligned}$$

which implies $c_1 + c_2 = 0$ and $c_1 = -(c_1 + 2c_2) \implies 2(c_1 + c_2) = 0 \implies c_1 + c_2 = 0$.

Hence, any $p(x) = c_0 + c_1x + c_2x^2$ where $c_1 + c_2 = 0$ works.

(b)

$$\begin{aligned} p(x) = p(1+x) &\implies c_0 + c_1x + c_2x^2 = c_0 + c_1(1+x) + c_2(1+x)^2 \\ &\implies c_0 + c_1x + c_2x^2 = c_0 + c_1 + c_1x + c_2(1+2x+x^2) \\ &\implies c_0 + c_1x + c_2x^2 = c_0 + c_1 + c_1x + c_2 + 2c_2x + c_2x^2 \\ &\implies c_0 + c_1x + c_2x^2 = (c_0 + c_1 + c_2) + (c_1 + 2c_2)x + c_2x^2 \end{aligned}$$

which implies $c_1 + c_2 = 0$ and $c_1 = c_1 + 2c_2 \implies c_2 = 0$. This in turn implies $c_1 = 0$.

Hence, only $p(x) = c_0$ works (but c_0 can be anything).

(c)

$$\begin{aligned} p(2x) = 2p(x) &\implies c_0 + c_1(2x) + c_2(2x)^2 = 2(c_0 + c_1x + c_2x^2) \\ &\implies c_0 + 2c_1x + 4c_2x^2 = 2c_0 + 2c_1x + 2c_2x^2 \end{aligned}$$

which implies $c_0 = 2c_0 \implies c_0 = 0$ and $4c_2 = 2c_2 \implies 2c_2 = 0 \implies c_2 = 0$

Hence, only $p(x) = c_1x$ works (but c_1 can be anything).

(d)

$$\begin{aligned}
 p(3x) = p(x+3) &\implies c_0 + c_1(3x) + c_2(3x)^2 = c_0 + c_1(x+3) + c_2(x+3)^2 \\
 &\implies c_0 + 3c_1x + 9c_2x^2 = c_0 + c_1x + 3c_1 + c_2(x^2 + 6x + 9) \\
 &\implies c_0 + 3c_1x + 9c_2x^2 = c_0 + c_1x + 3c_1 + c_2x^2 + 6c_2x + 9c_2 \\
 &\implies c_0 + 3c_1x + 9c_2x^2 = (c_0 + 3c_1 + 9c_2) + (c_1 + 6c_2)x + c_2x^2
 \end{aligned}$$

which implies $c_0 = c_0 + 3c_1 + 9c_2 \implies c_1 + 3c_2 = 0$, $3c_1 = c_1 + 6c_2 \implies c_1 = 3c_2$, and $9c_2 = c_2 \implies 8c_2 = 0 \implies c_2 = 0$. But this in turn implies that $c_1 = 0$. Thus, only $p(x) = c_0$ works (but c_0 can be anything).

□

Problem 12

Solution:

(a) By the binomial theorem

$$(1+x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^k (1)^{2n-k} = \sum_{k=0}^{2n} \binom{2n}{k} x^k = \sum_{k=0}^{2n} c_k x^k$$

where $c_k = \binom{2n}{k}$. Hence, we have a polynomial of degree $m = 2n$

(b) Observe that

$$1 - x^{n+1} = 1^{n+1} - x^{n+1} = (1-x)(x^n + x^{n-1}(1) + \cdots + x^0(1)) = (1-x)(x^n + x^{n-1} + \cdots + x + 1)$$

so we have

$$\frac{1 - x^{n+1}}{1 - x} = x^n + x^{n-1} + \cdots + x + 1 = \sum_{k=0}^n c_k x^k$$

where $c_k = 1$. Hence, we have a polynomial of degree n .

(c) Working out a few cases illustrates a pattern

$$\begin{aligned}
 \prod_{k=0}^0 (1 + x^{2^k}) &= (1 + x) \\
 \prod_{k=0}^1 (1 + x^{2^k}) &= (1 + x)(1 + x^2) = 1 + x + x^2 + x^3
 \end{aligned}$$

$$\prod_{k=0}^2 (1+x^{2^k}) = (1+x)(1+x^2)(1+x^4) = (1+x+x^2+x^3)(1+x^4) = 1+x+x^2+x^3+x^4+x^5+x^6+x^7$$

We will prove by induction that

$$\prod_{k=0}^n (1+x^{2^k}) = \sum_{k=0}^{2^{n+1}-1} x^k$$

i.e. we have a polynomial of degree $2^{n+1} - 1$. For the base case ($n = 0$) observe that

$$\prod_{k=0}^0 (1+x^{2^k}) = (1+x) = \sum_{k=0}^1 x^k = \sum_{k=0}^{2^1-1} x^k = \sum_{k=0}^{2^1-1} x^k = \sum_{k=0}^{2^{0+1}-1} x^k$$

For the inductive hypothesis, assume that for arbitrary fixed $0 \leq j < n$, we have

$$\prod_{k=0}^j (1+x^{2^k}) = \sum_{k=0}^{2^{j+1}-1} x^k$$

Now, for the inductive step, observe that

$$\begin{aligned}
 \prod_{k=0}^n (1 + x^{2^k}) &= (1 + x^{2^n}) \prod_{k=0}^{n-1} (1 + x^{2^k}) \\
 &= (1 + x^{2^n}) \sum_{k=0}^{2^{(n-1)+1}-1} x^k && \text{(by inductive hypothesis)} \\
 &= (1 + x^{2^n}) \sum_{k=0}^{2^n-1} x^k \\
 &= \sum_{k=0}^{2^n-1} x^k + \sum_{k=0}^{2^n-1} x^k x^{2^n} \\
 &= \sum_{k=0}^{2^n-1} x^k + \sum_{k=0}^{2^n-1} x^{2^n+k} \\
 &= \sum_{k=0}^{2^n-1} x^k + \sum_{k=2^n}^{2^n+2^n-1} x^k \\
 &= \sum_{k=0}^{2^n-1} x^k + \sum_{k=2^n}^{2(2^n)-1} x^k \\
 &= \sum_{k=0}^{2^n-1} x^k + \sum_{k=2^n}^{2^{n+1}-1} x^k \\
 &= \sum_{k=0}^{2^{n+1}-1} x^k
 \end{aligned}$$

□